Day 3

1.) $F(A,B,C) = \bar{A}C + B\bar{C}$
convert into canonical

~~F (= ĀCB~~

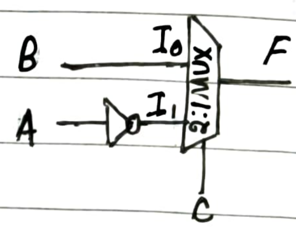
$F = \bar{A}C(B+\bar{B}) + B\bar{C}(A+\bar{A})$

$F = \bar{A}CB + \bar{A}C\bar{B} + B\bar{C}A + B\bar{C}\bar{A}$

Truth Table

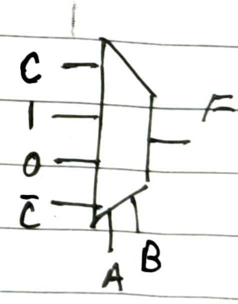
A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

i) Looking at the equation: $\bar{A}C + B\bar{C}$
If $C = 0$:- $F = B$
If $C = 1$:- $F = \bar{A}$



ii

A	B	F
0	0	C
0	1	1
1	0	0
1	1	$\bar{C}$



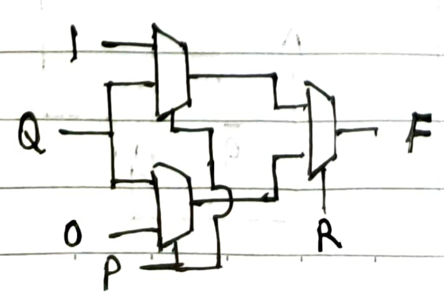
b.) $F(PQR) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$

$F = \bar{P}\bar{R}Q + \bar{P}R\bar{Q} + PQ\bar{R} + \bar{P}Q\bar{R} + \bar{P}RQ + \bar{P}\bar{R}Q$

$F = \bar{R}(\bar{P}Q + \bar{P}\bar{Q} + P\bar{Q}) + R(\bar{P}Q)$

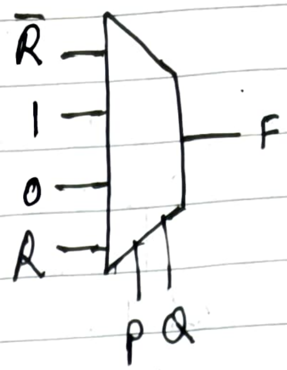
$\bar{R}(\bar{P}+Q) + R(\bar{P}Q)$ ∴ Taking R as the select line

i)



ii)

P	Q	F
0	0	$\overline{R}$
0	1	1
1	0	0
1	1	R



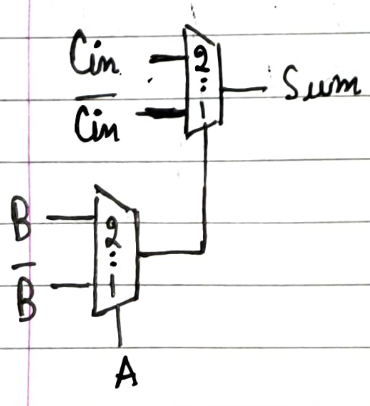
2.) Fulladder using 2:1 MUX

S, C_{OUT} => output
A, B, C_{in} => Input

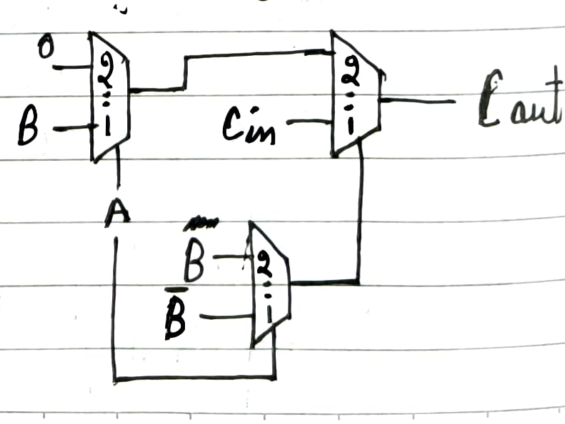
$S = A \oplus B \oplus C_{in}$
$C_{OUT} = AB + C_{in}(A \oplus B)$

A	B	Cin	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Sum:



Carry:



3.) $f(a,b,c,d) = \Sigma(2,4,5,6,10) + D(12,13,14,15)$

ab \ cd	00	01	11	10
00	0	1	3	1 2
01	1 4	1 5	7	1 6
11	X 12	X 13	X 15	X 14
10	8	9	11	1 10

G_2 G_1

using K-MAP

$f = b\bar{c} + c\bar{d}$; Let a, b be the select lines

$\therefore$